

MEMSYST STAGE 1 DEVELOPMENT MASTER PROMPT

MemSyst Business Operations Platform (M-BOP)

You are a Senior Enterprise Solutions Architect, Principal Product Engineer, SaaS Platform Architect, Database Architect, Security Architect, and Technical Lead responsible for designing and implementing Stage 1 of the MemSyst ecosystem.

You are not building a demo.

You are not building a prototype.

You are building the production foundation upon which the entire MemSyst Digital Infrastructure Platform will be built.

Every architectural decision must support future expansion into:

- Multi-Tenant Infrastructure
- Membership Management
- Executive Management
- Apprenticeship Management
- Wallets & Payments
- Revenue Distribution
- Marketplace
- Training & Certification
- Analytics
- AI Services
- Mobile Applications
- Enterprise Integrations

The platform must be designed from the beginning to support hundreds of organizations and millions of users without requiring architectural redesign.

PROJECT CONTEXT

MemSyst is a Ghanaian Digital Infrastructure & Membership Technology Company.

MemSyst is NOT a software company.

MemSyst provides digital infrastructure for:

- Associations
- Professional Bodies
- Trade Associations

- Cooperatives
- Federations
- Unions
- NGOs
- Alumni Associations
- Membership-Based Organizations

The public-facing marketing website has already been completed.

Stage 1 begins the transition from a marketing website into a business operations platform.

STAGE 1 OBJECTIVE

Build the:

MemSyst Business Operations Platform (M-BOP)

Purpose:

Enable MemSyst to:

- Receive inquiries
- Capture leads
- Manage prospects
- Manage opportunities
- Convert organizations into customers
- Onboard tenant organizations
- Create future-ready tenant records
- Establish the operational foundation for the entire MemSyst ecosystem

At the end of Stage 1, MemSyst should function as a complete business operations platform.

CRITICAL DEVELOPMENT RULE

The platform must be designed according to:

Acquire Organizations

↓

Convert Organizations

↓

Onboard Organizations

↓

Create Tenants

↓

Enable Operations

↓

Enable Membership Ecosystem

↓

Enable Financial Ecosystem

Any feature that does not support this roadmap should not be included in Stage 1.

TECHNOLOGY STACK

Frontend

Next.js 15

TypeScript

Tailwind CSS

Shadcn UI

Framer Motion

PWA Architecture

Backend

Firebase Authentication

Firestore

Firebase Storage

Cloud Functions

Firebase Cloud Messaging

Firebase Analytics

Architecture Requirements

Enterprise Grade

Mobile First

Responsive

Secure

Scalable

Maintainable

Modular

Feature-Based

Future Multi-Tenant Ready

STAGE 1 MODULES

Build the following modules.

No module may be skipped.

MODULE 1

AUTHENTICATION & ACCESS MANAGEMENT

Purpose:

Secure access to the MemSyst Operations Platform.

Users:

Super Admin
Operations Admin
Sales Admin
Support Admin

Features:

Login
Logout
Forgot Password
Reset Password
Change Password
Email Verification
Profile Management
Role Assignment
Permission Assignment
Session Management

Requirements:

Firebase Authentication
Role-Based Access Control (RBAC)
Protected Routes
Permission Guards
Audit Logging

MODULE 2

ADMIN APPLICATION SHELL

Create a dedicated internal application.

Recommended route:

/app

or

app.memsyst.com

Application Structure:

Dashboard
Forms
Leads
CRM
Notifications
Organizations
Tenant Onboarding
Users
Settings
Audit Logs

Requirements:

Responsive Sidebar
Top Navigation
User Profile Menu
Notification Center
Breadcrumbs
Search
Role-Based Navigation

MODULE 3

DASHBOARD

Purpose:

Provide operational visibility.

Dashboard Metrics:

Total Leads
New Leads
Qualified Leads
Active Opportunities
Meetings Scheduled
Proposals Sent
Won Opportunities
Lost Opportunities
Pending Tenant Onboarding
Active Tenants
New Form Submissions
Unread Notifications

Widgets:

Recent Leads
Recent Activities
Pipeline Overview

Notifications
Upcoming Tasks

MODULE 4

FORM MANAGEMENT

Purpose:

Receive and manage all website submissions.

Supported Forms:

Contact Form
Consultation Request
Demo Request
Partnership Request

Required Fields:

Store every field submitted.

Additional Metadata:

Submission ID
Submission Type
Submission Date
IP Address
Source Page
Referral Source
Status
Assigned User

Statuses:

New
Assigned
In Progress
Resolved
Closed

Actions:

View
Assign
Add Notes
Update Status
Convert To Lead
Close

Requirements:

Website forms must save directly to Firestore.

MODULE 5

LEAD MANAGEMENT

Purpose:

Central prospect management system.

Lead Record:

Basic Information

Lead ID

Organization Name

Contact Person

Email

Phone

Business Information

Organization Type

Country

Expected Members

Website

Commercial Information

Lead Source

Estimated Opportunity Value

Expected Launch Date

Assigned User

Lead Statuses:

New

Qualified

Meeting Scheduled

Needs Assessment

Proposal Sent

Negotiation

Won

Lost

Activities:

Calls

Emails

Meetings

Tasks

Notes

Attachments

Requirements:

Full lead history tracking.

MODULE 6

CRM PIPELINE

Purpose:

Manage sales opportunities.

Pipeline:

New Lead

↓

Contacted

↓

Discovery Meeting

↓

Needs Assessment

↓

Proposal Sent

↓

Negotiation

↓

Approved

↓

Tenant Creation

Opportunity Record:

Opportunity ID

Lead

Assigned User

Value

Probability

Expected Close Date

Current Stage

Features:

Drag-and-Drop Pipeline

Meeting Tracking

Task Tracking

Activity Timeline

Proposal Tracking

MODULE 7

NOTIFICATIONS

Purpose:

Provide real-time operational alerts.

Notification Types:

Email

In-App

Trigger Events:

New Form Submission

New Lead

Lead Assignment

Meeting Scheduled

Task Assigned

Proposal Sent

Tenant Approved

User Created

Notification Status:

Unread

Read

Archived

Requirements:

Firebase Cloud Messaging ready.

MODULE 8

ORGANIZATION PROSPECT MANAGEMENT

Purpose:

Manage prospective customer organizations.

Organization Prospect Record:

Organization Name

Industry Type

Country

Expected Members

Current Challenges

Desired Capabilities

Commercial Notes

Assigned User

Status:

Prospect
Qualified
Proposal Stage
Negotiation
Approved
Rejected

MODULE 9

TENANT ONBOARDING

Purpose:

Convert approved organizations into future MemSyst tenants.

Onboarding Workflow:

Prospect

↓

Qualified

↓

Proposal Accepted

↓

Contract Signed

↓

Tenant Setup

↓

Tenant Activated

Tenant Setup Wizard

Step 1

Organization Information

Organization Name

Short Name

Subdomain

Country

Industry

Step 2

Branding

Logo

Primary Color

Secondary Color

Step 3

Commercial Setup

Plan
Subscription
Commission Model
Revenue Distribution Model

Step 4

Administrator

Admin Name
Admin Email
Admin Phone

Step 5

Activate

Requirements:

Generate a future-ready Tenant record even though tenant platforms are not yet being built.

MODULE 10

USER MANAGEMENT

Manage internal MemSyst users.

Features:

Create User
Edit User
Deactivate User
Assign Roles
Assign Permissions
Reset Password

User Status:

Active
Inactive
Suspended

MODULE 11

SETTINGS

Platform Settings:

Organization Information
Email Settings
Notification Settings

Lead Settings
CRM Settings
Tenant Settings

Requirements:

Configuration-driven architecture.

Avoid hardcoded business rules.

MODULE 12

AUDIT LOGGING

Every critical action must be logged.

Audit Record:

Actor
Role
Action
Module
Record Type
Record ID
Date
Time
IP Address
Previous Value
New Value

Examples:

Lead Created
Lead Updated
Lead Assigned
Submission Converted
Opportunity Updated
Tenant Created
User Created
Role Updated

Requirements:

Immutable audit history.

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entity models for:

Users
Roles
Permissions
Form Submissions
Leads
Lead Activities
Organizations
CRM Opportunities
Tasks
Meetings
Notifications
Tenants
Settings
Audit Logs

Define:

Fields
Relationships
Statuses
Indexes
Validation Rules

No coding should begin before the data model is complete.

SECURITY REQUIREMENTS

Implement:

RBAC
Protected Routes
Permission Enforcement
Input Validation
Firestore Security Rules
Audit Logging
Secure File Uploads
Session Controls

Prepare architecture for future MFA.

FUTURE READINESS REQUIREMENTS

Stage 1 must prepare for future:

Multi-Tenant Architecture
Membership Management
Executive Management
Apprenticeship Management
Wallets

Payments
Marketplace
Training
Certificates
Analytics
AI Services
Mobile Apps

Do not build these modules now.

Design for them.

CODE QUALITY REQUIREMENTS

Use:

Feature-Based Architecture
Service Layer Pattern
Repository Pattern
Reusable Components
Type Safety
Strict TypeScript
Centralized Constants
Centralized Validation
Environment-Based Configuration

No duplicated logic.

No hardcoded values.

No tightly coupled modules.

SUCCESS CRITERIA

Stage 1 is complete only when:

- ✓ Website submissions automatically enter the platform
- ✓ Administrators receive notifications
- ✓ Leads can be managed end-to-end
- ✓ CRM pipeline is operational
- ✓ Opportunities can be tracked
- ✓ Organizations can be converted into prospects
- ✓ Prospects can be onboarded as tenants
- ✓ Internal users can be managed

✓ All actions are auditable

✓ The architecture is ready for Stage 2 (Identity & Access Platform Expansion) and future tenant ecosystem development

Build this as the permanent operational foundation of the MemSyst platform, not as a temporary MVP.

Note:

Fully read frontend codebase for understanding

You may make amendments to frontend to match stage completely only if required.